Corrected and conduct of chapter 7

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. 0. *Action:* Take over the criterion of orthogonality.

2. Yes: 2 $-2=0$. *Action:* Replace x and y simultaneously.

3. $x=3$, $y=2$. *Action:* Add and replace.

4. 5. *Action:* Calculate the vector and then its norm.

5. $(x-2)^2-4$. *Action:* Develop to check.

6. The only torque (0; 0). *Action:* Use the sum of two positive squares.

Research activity

Trials suggest H(2;4), but Pythagoras in PHM demonstrates minimality.
with normal (2;-1).

Immediate applications

1. $3(x-2)+4(y+1)=0$, or 3 $x+4y-2=0$.

2. H(2; 1) and distance 2; normal is horizontal.

3. $(x+1)^2+(y-4)^2=0$: the only point (-1; 4), not a strictly positive radius circle.

4. $(x-1)^2=0$, so only point (1; 1); the right is tangent to the circle.

Exercises: approach and results

Exercise 7.1 · Build a straight line. $2(x+1)+5(y-3)=0$, or 2 $x+5y-13=0$.
$(5;-2)$, therefore B(4; 1) is on the right; control 8 + 5 $-13=0$.

Exercise 7.2 · Calculate a project. Normal (2; $-1)$; $\lambda=\dfrac{8-3}{5}=1$. $H=(4;3)-(2;-1)=(2;4)$. $2(2)-4=0$ and $\overrightarrow{PH}=(-2;1)$ is normal; distance $\sqrt{5}$.

Exercise 7.3 · Recognize a circle. $(x-3)^2+(y+2)^2=25$; centre (3; $-(2)$, section 5.
$y=-2$, $x=3\pm5$: $(-2;-(2)$ and $(8;-2)$.

Exercise 7.4 · Mediator. Environment I(1;2), normal $\overrightarrow{AB}=(4;2)$. $4(x-1)+2(y-2)=0$, or $2x+y-4=0$. I check 2 + 2 $-4=0$.

Exercise 7.5 · The best access. Projected H(2;4) minimizes PH, length $\sqrt{5}$ km. For all M on the road, the PHM triangle is right-angled at H and $PM^2=PH^2+HM^2\geq PH^2$. Equality for M=H.

Exercise 7.6 · Parameter and set. $(x-1)^2+(y+2)^2=5-k$. If $k<5$, center circle $(1;-2)$ and radius $\sqrt{5-k}$; if k = 5, point; if $k>$ Five, empty.

Exercise 7.7 · A parabola as a locus. $MF^2=x^2+(y-2)^2$, distance to square axis y^2. Equality: $x^2-4y+4=0$, or $y=\frac{x^2}{4}+1$. Positive distances allow for mutual exchange.

Exercise 7.8 · Deferred review. Centre (2; 0), radius 2 ; ($x-2)^2+y^2$ = 4. For x=5, $y^2=-5$, Impossible. Interpretation: Right at distance 3 from the centre, above radius.

CQFD, Exercise 18 p. 279, questions a, b, c and d. Possible responses are ($-1;2)$, $(3;-6)$, $(\frac{1}{4};-1)$ and (1; 0). Any non-zero multiple is suitable. x = 4, write $x-4=0$.

CQFD, Exercise 21 p. 279, questions a and b. a : $x-(y-2)=0$, or $x-y+2=0$. b : $4(x+1)+6(y-3)=0$, or 2 $x+3y-7=0$. Substitute coordinates of points confirms membership.

MCQ: answers and reasons

MCQ 1.

1. B. X and y coefficients.
2. A. Product with normal zero.
3. D. Read the coordinates opposite the signs.
4. C. Positive root of 9.
5. B. Sum of non-negative squares.

MCQ 2.

1. C. Same abscissa, ordered null.
2. A. Absolute vertical deviation.
3. D. Non-zero normal.
4. B. The two squares shall be void.
5. C. $y^2=0$.

True or false

1. True: per P passes a single perpendicular to the right; if P belongs to the right, H=P.
2. False: if c=0, one dot; if c>0, empty set.
3. False: substitution of a right parameter in the circle equation gives an equation quadratic with strictly positive dominant coefficient, therefore at most two roots.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.